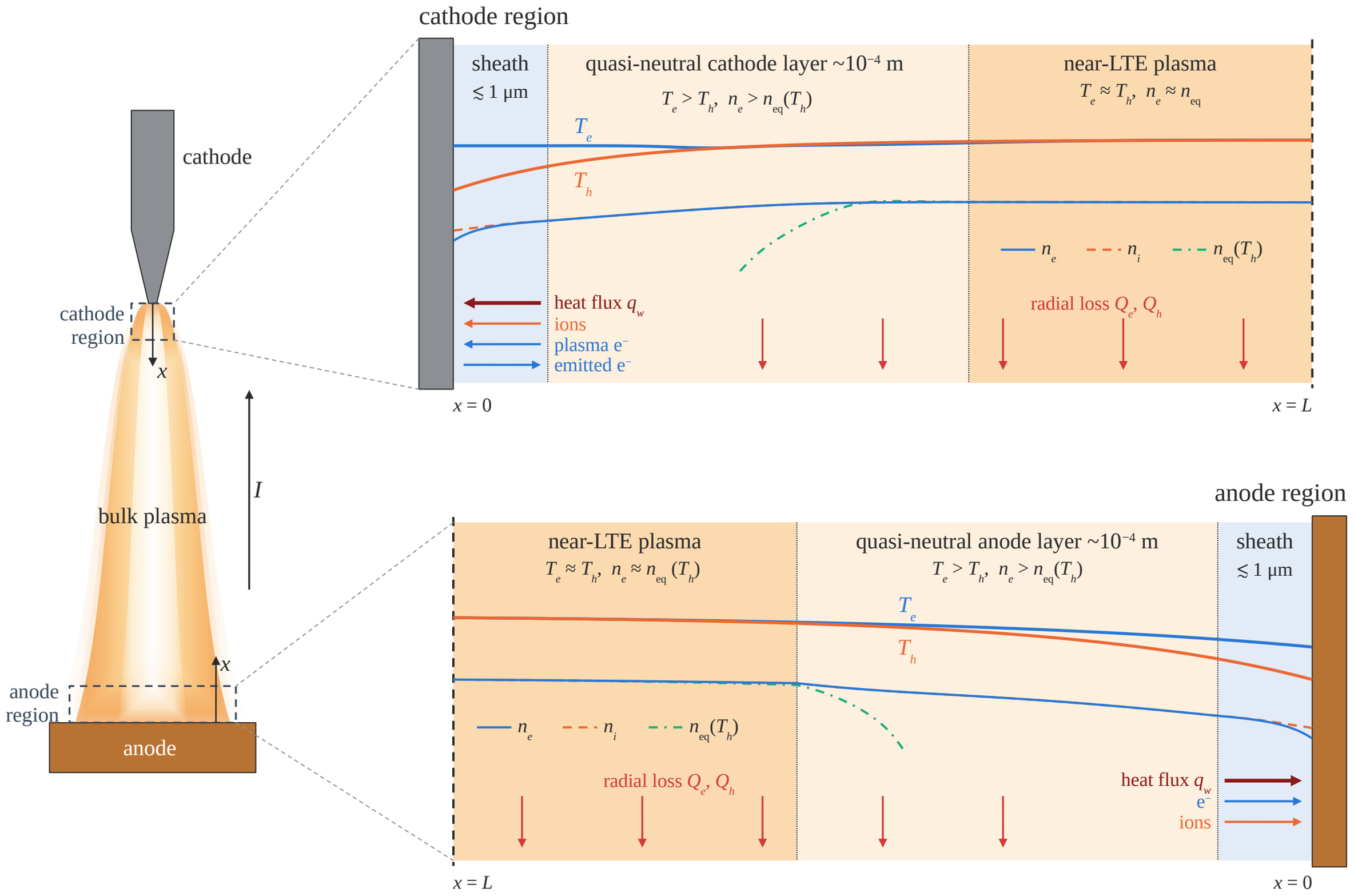

cathode region
sheath
≲ 1 μm
quasi-neutral cathode layer ~10⁻⁴ m
$T_e > T_h,\ n_e > n_{eq}(T_h)$
near-LTE plasma
$T_e \approx T_h,\ n_e \approx n_{eq}$
T_e
T_h
n_e n_i $n_{eq}(T_h)$
heat flux q_w
ions
plasma e⁻
emitted e⁻
radial loss Q_e, Q_h
$x = 0$
$x = L$
cathode
cathode region
x
I
bulk plasma
anode region
near-LTE plasma
$T_e \approx T_h,\ n_e \approx n_{eq}(T_h)$
quasi-neutral anode layer ~10⁻⁴ m
$T_e > T_h,\ n_e > n_{eq}(T_h)$
sheath
≲ 1 μm
T_e
T_h
n_e n_i $n_{eq}(T_h)$
radial loss Q_e, Q_h
heat flux q_w
e⁻
ions
x
anode region
anode
$x = L$
$x = 0$